Longitudinal Data Analysis

Homework 2

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December 2024

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1 Introduction

Chronic kidney disease is a significant public health problem and one of the leading causes of death associated with non-communicable diseases. In 2017, approximately 1.2 million deaths were attributed to CKD, with a global prevalence of 9.1%, reflecting a 29.3% increase since 1990. End-Stage Renal Disease (ESRD) is the most severe form of CKD, characterized by a substantial decline in kidney function that requires interventions such as hemodialysis or kidney transplantation to sustain life .

Patients with ESRD frequently experience complications, including anemia, primarily due to the kidneys' reduced ability to produce erythropoietin (EPO). EPO is a crucial hormone that stimulates red blood cell production (erythropoiesis) in the bone marrow, and its deficiency leads to reduced hemoglobin levels. Additionally, hemodialysis can further reduce iron stores, exacerbating anemia. In severe cases, anemia can result in cardiovascular complications, significantly affecting the overall health and quality of life of these patients.

To effectively manage anemia in patients with ESRD, a comprehensive approach is essential. The current strategy primarily includes the administration of recombinant EPO. Since adequate iron levels are necessary for erythropoiesis, it is crucial to evaluate and maintain sufficient iron stores. Accordingly, previous studies have shown that iron deficiency can reduce the effectiveness of EPO, leading to persistent anemia despite treatment. Nevertheless, some patients do not respond to EPO therapy even when adequate iron stores are present. Therefore, this study aims to investigate how changes in hemoglobin levels over time are influenced by EPO dosing, iron status, and demographic factors, providing insights to improve anemia management in patients with ESRD receiving hemodialysis.

1.1 Objectives

This study aims to investigate how hemoglobin (Hb) status, categorized as severe anemia, mild anemia, and normal hemoglobin, changes over time among patients with renal deficiency undergoing hemodialysis. Specifically, we aim to evaluate the association between the dose of EPO administered and transitions between these categories. Additionally, the study will assess the influence of iron deficiency status and demographic factors, including age and sex, on these transitions.

2 Data Management and Exploration

2.1 Data Management

The dataset used for this study includes monthly measurements of hemoglobin (Hb, categorized as follows: **severe anemia** if Hb ≤ 10 g/dL, **mild anemia** if $10 <$ Hb ≤ 12 g/dL, and **normal** if Hb > 12 g/dL), EPO dose (IU/kg/week, IU: International Unit, discrete), and iron deficiency status (categorical, defined as present if serum ferritin levels are below 100 μg/L and/or transferrin saturation is below 20%). Additionally, demographic data,

including sex (categorical) and age (discrete), were collected.

The dataset contained data from 3823 patients receiving hemodialysis. The total follow-up period lasted 6 months. However, not all patients attended every visit and a large number were lost to follow-up after the 4th month, resulting in an unbalanced dataset. On average patients attended approximately 2 to 3 visits. Table 1 summarizes the number of patients according to their visit counts.

During the follow-up, patients received EPO treatment to recover from anemia and maintain hemoglobin levels within the desired range, with hemoglobin measurements used to adjust the EPO dose for the following month. Given that there is no immediate biological effect of EPO on hemoglobin, and the EPO dose is adjusted according to current hemoglobin levels, further analysis might reflect clinicians' decisions rather than the true effect of EPO on hemoglobin over time. Therefore, to better capture the effect of EPO on transitions between hemoglobin categories (HBCat), we generated two variables: lagged dose and lagged Hb.

Lagged dose represents the EPO dose received in the previous month. Since the effect of EPO on hemoglobin levels typically becomes evident, this variable accounts for the delayed effect of EPO on hemoglobin, allowing for a more accurate analysis of the relationship between prior EPO doses and subsequent changes in hemoglobin. Lagged hemoglobin (lagHb) represents the hemoglobin measurement from the previous month. It accounts for the temporal stability of hemoglobin levels and helps explain transitions between hemoglobin categories (HbCat) over time. Additionally, lagHb reflects the hemoglobin levels used by clinicians to adjust EPO dosing, providing crucial context for understanding subsequent hemoglobin outcomes.

Iron deficiency was analyzed as observed during the same month rather than lagged, as it reflects the iron status from the prior month. Intravenous iron therapy, routinely administered to hemodialysis patients with iron deficiency, typically increases hemoglobin levels within one to two weeks, repleting the iron stores within days. This rapid response supports the assumption that the recorded iron deficiency reflects prior status at the time of treatment adjustment.

Table 1: Number of Patients by Visit Count

Visits	1	2	3	4	5	6
Number of Patients	535	828	1180	780	127	28

2.2 Exploratory Data Analysis

The exploratory analysis included individual longitudinal profiles of Hb categories to visualize patient's transitions across Hb categories over time. For better visualization of the patterns in the data, we randomly sampled 20 patients from the population. As shown in Figure 1, these profiles revealed considerable variability. Some patients remained in the stable in the same Hb category, while others showed a non-monotonic evolution, recovering and regressing or worsening before improving again. This non-linear patter

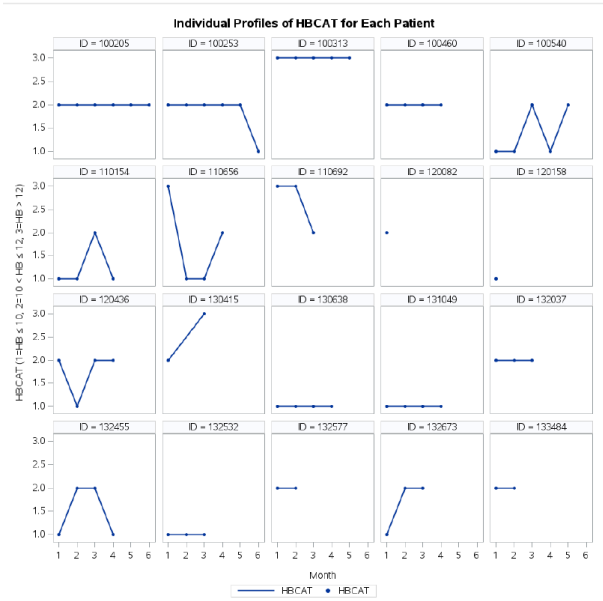

Figure 1: Individual Profiles of HBCAT over Months

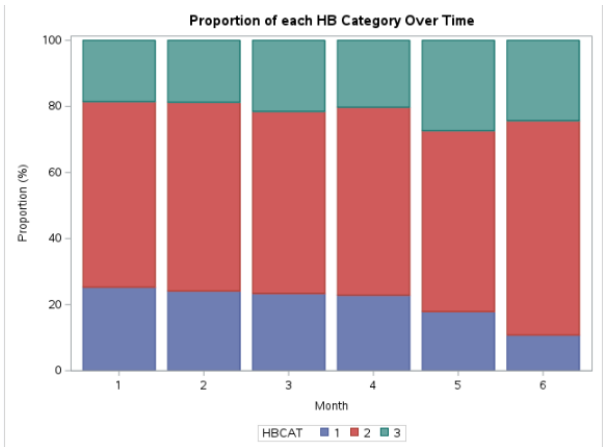

Figure 2: Proportion of HB Categories per Month

of transitions likely reflects the dynamic adjustments in EPO doses and changes in iron deficiency status over time, suggesting the need to account for their interaction with temporal trends in the analysis.

To complement the individual profiles, we plotted the proportions of patients in each HBCAT category per month (Figure 2). The plot shows highlights the relative stability in the distribution of hemoglobin categories over time, with mild anemia (HBCAT=2) consistently the most common category. However, subtle shifts are evident, including an increase in the proportion of patients with normal hemoglobin (HBCAT=3) up to Month 5, followed by a slight decline in Month 6.

We plotted the distribution of our main predictor, LagDose, within HB categories over time (Figure 3). The boxplots show that patients with severe anemia) consistently received higher doses of EPO compared to those with mild anemia and normal Hb, reflecting treatment intensification for severe anemia. Interestingly, the distributions for

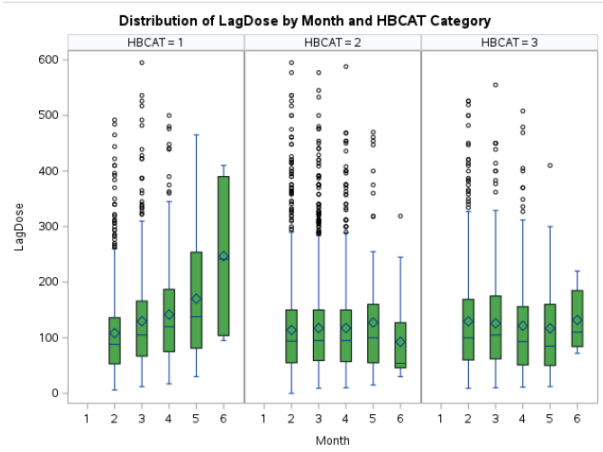

Figure 3: Distribution of Lagged Epo Dose by Month and Hb category

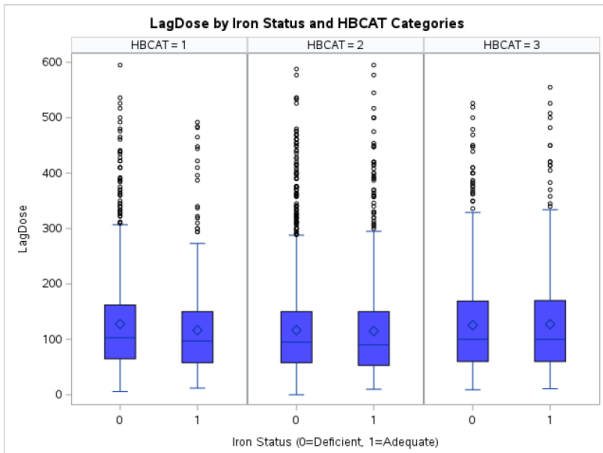

Figure 4: Distribution of LagDose by Ion Status within Hb Categories

mild anemia and normal Hb categories overlap, suggesting that dosing strategies may not differ substantially between these categories. However, this plot assumes that patients remain in the same HB category over time, which does not account for transitions between categories or repeated measures. These limitations highlight the need for longitudinal models to explore how EPO dosing influences transitions and hemoglobin changes across categories.

Figure 4 shows the distribution of lagdose by iron status within HB categories. Patients with severe anemia consistently received higher and more variable doses of EPO, with iron-deficient patients tending to receive slightly higher doses than iron-adequate patients, reflecting the clinical challenge of managing anemia with iron deficiency. For mild anemia and normal Hb, the LagDose distributions overlap between iron-deficient and iron-adequate groups, suggesting that iron status plays a less prominent role in dosing decisions as hemoglobin levels improve. These patterns suggest that the effect of EPO dosing on hemoglobin categories may be modified by iron status, particularly in severe anemia, and support the potential inclusion of interaction terms.

We explored the interaction between LagHb and Iron Status to assess how iron deficiency affects the relationship between previous hemoglobin levels and future hemoglobin response. Since iron deficiency can reduce the effectiveness of EPO, understanding this

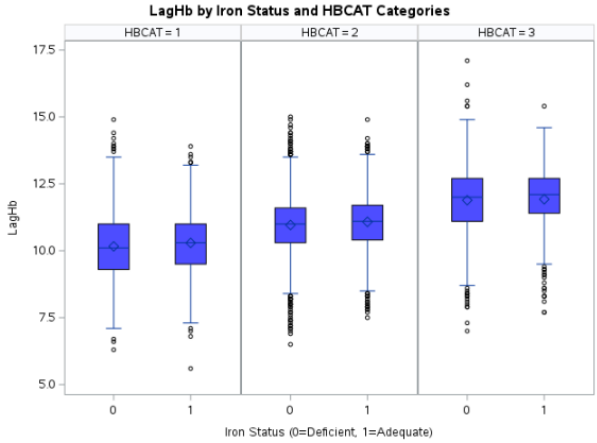

Figure 5: Distribution of LagHb by Iron Status within Hb Categories

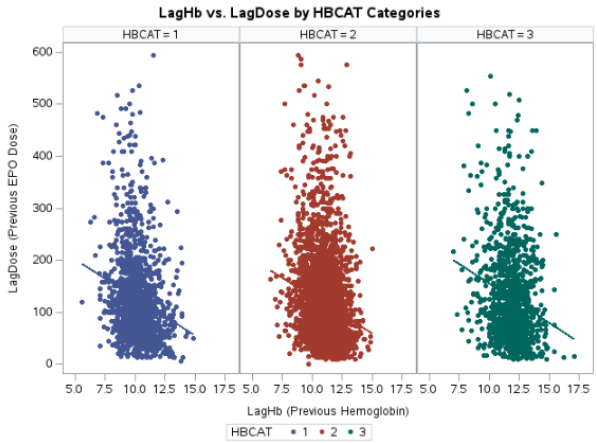

Figure 6: Scatterplot of LagHb vs LagDose by Hb category

interaction is critical for EPO dosing. Figure 5 shows that there is not much variability in LagHb between iron-deficient and iron-adequate patients, and within all three HB categories.

Given that EPO dosing is adjusted based on LagHb, we also examined the interaction between LagDose and LagHb to see how previous hemoglobin levels influence subsequent dosing. Figure 6 clearly shows a negative relationship between LagHb and LagDose, confirming that higher LagHb levels are associated with lower EPO doses, especially in severe anemia. This supports the clinical approach of adjusting EPO doses based on hemoglobin levels.

Cumulative logit plots were used to examine how transitions between the HB categories are influenced by Month, LagDose, LagHb and Iron status over time. These plots show how the cumulative log odds of each Hb category change as these factors vary. Briefly, Figure 7 evidences non-parallel cumulative log odds for the effect of month (time) suggesting a violation of the proportional odds assumption. Figure 8, evidences non-parallel cumulative log odds slopes for the effect of lagdose, also suggesting a violation of the proportional odds assumption, and possible non linear relationship due to variability in logit 2. Figure 9, evidences non-parallel cumulative log odds slopes for the effect of laghb and variability also in logit2, also suggesting a violation of the proportional odds assumption, and possible non linear relationship. Lastly, Figure 10, evidences clearly

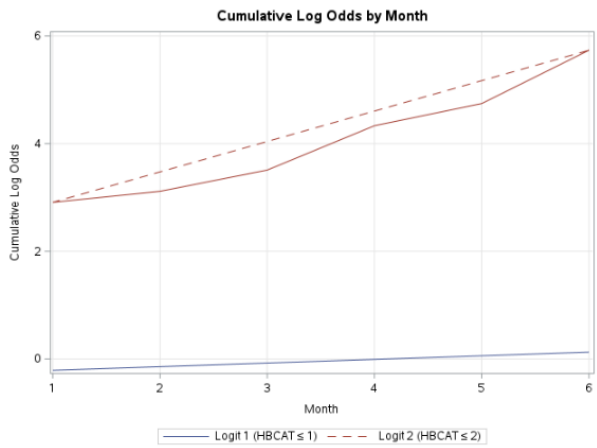

Figure 7: Cumulative Log Odds over Month

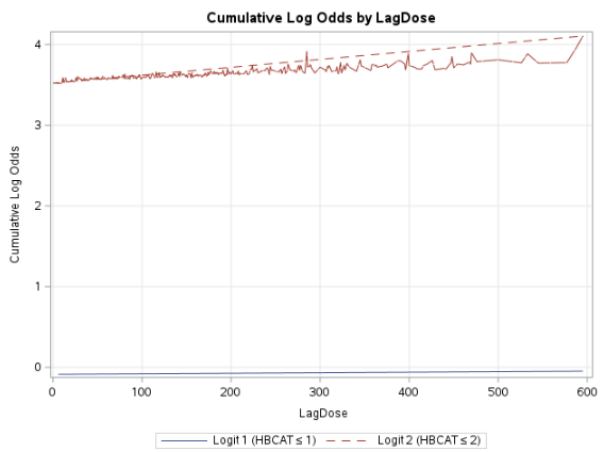

Figure 8: Cumulative Log Odds over LagDose

Figure 9: Cumulative Log Odds over LagHb

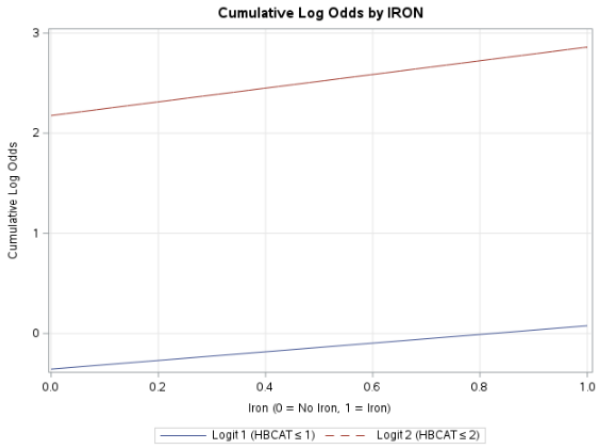

Figure 10: Cumulative Log Odds over Iron

parallel cumulative log odds slopes for the effect of iron, suggesting that the proportional odds assumption holds.

3 Material and Methods

3.1 Proportional Odds Assumption

For an ordinal response Y with K ordered categories, the modeling focus is on predict the cumulative probabilities of being in a given category or lower ($PY <= k$), rather than directly modeling the probability of response, connecting the ($PY <= k$) to the linear predictor through a link function, typically the logit link,

To simplify interpretation and make the model more parsimonious, the effects of the regression coefficients can be assumed not to vary across thresholds, therefore assuming the Proportional Odds Logistic Regression (POLR) model:

$$\text{logit}(P(Y \leq k)) = \eta_k = \alpha_k + X\beta, \quad k = 1, \ldots, K-1$$

where

- α_k: Threshold-specific intercept
- $X\beta$: Linear predictor constant accross all thresholds

POLR uses the cumulative logit given by:

$$\text{logit}(P(Y \leq k)) = \log\left(\frac{1 - P(Y \leq k)}{P(Y \leq k)}\right)$$

3.1.1 GEEORD macro approach

Cumulative log odds graphs, while useful, have limitations such as computational demands or subjective visual interpretation. The GEEORD macro, automated in SAS, addresses these issues by providing a more structured and efficient approach to model specification and assumption testing, specifically in the context of longitudinal ordinal

data. This macro leverages the Generalized Estimating Equations framework to handle within-subject correlations that exist in repeated measures data. GEEORD allows us to fit both proportional odds models and partial proportional odds models, making it thus well-suited for datasets where the proportional odds assumption may not hold for all predictors. The macro conducts assumption testing using global score tests and variable-specific Wald tests, providing robust statistical evidence regarding whether the proportional odds assumption is valid. Additionally, GEEORD can incorporate various working correlation structures to account for the dependencies between repeated measurements.

Taking into account the previous visual findings, further testing specifically for the Lagged_dose variable with a Wald test proved that the proportionality assumption does not hold. The case was similar for the Lagged_Hb variable which already showed clear evidence of violation.

It is to be noted however that GEEORD it has limitations as well. It assumes that missing data are Missing Completely at Random, which may not always be the case. For assumption testing, GEEORD's reliance on score and Wald tests can sometimes yield results that are difficult to interpret, especially when small sample sizes or rare outcome categories are involved.

3.1.2 Partial Proportional Odds

When the proportional odds assumption does not hold, alternatives like the partial proportional odds model or multinomial logistic regression can be explored to allow for greater flexibility in modeling the mean structure.

The Partial Proportional Odds (PPOLR) model, which relaxes the assumption for specific predictors while retaining it for others provides is specified with;

$$\text{logit}(P(Y \leq k)) = \alpha_k + X_1\beta + X_2\beta_k, \quad k = 1, \ldots, K-1$$

where:

- X_1: Predictors with constant β (proportional)
- X_2: Predictors with threshold-specific coefficients β_k (non-proportional)

Both marginal models (GEE) and hierarchical models (GLMM) can incorporate the POLR or PPOLR frameworks.

3.2 Generalized Estimating Equations (GEE)

+ To model the longitudinal transitions of hemoglobin categories (severe anemia, mild anemia, and normal hemoglobin) over time, we fitted a marginal model using Generalized Estimating Equations (GEE). This approach estimates the population-averaged effects of predictors on transitions between the three hemoglobin categories, accounting for within-subject correlation due to repeated measures. GEE employs a quasi-likelihood method to estimate model parameters, which does not require strict assumptions about

the distribution of the outcome data. Instead of relying on a fully specified likelihood function, the quasi-likelihood approach focuses on estimating parameters in a way that remains robust to model misspecifications. GEE allows the specification of a working correlation structure to model within-subject dependence, ensuring reliable estimates even when the correlation structure is misspecified or there is missing data. Therefore, this method is particularly suitable for our dataset, providing robust standard errors even in the presence of unbalanced data and dropout after Month 4.

Given the ordinal nature of the response variable (Hb categories), we initially considered a cumulative logistic regression model, also known as Proportional Odds Logistic Regression. This model estimates the log odds of being in a higher category relative to lower categories (severe vs mild and normal, severe and mild vs normal) and assumes that the effect of predictors is consistent across all ordinal categories. However, exploratory analysis revealed violations of the proportional odds assumption for several predictors, including LagDose, LagHb, and Month. Specifically, the cumulative log odds plots were not parallel across the ordinal thresholds, suggesting that the effects of these predictors vary at different points in the ordinal scale. To address this, we employed a Partial Proportional Odds (PPOLR) model, which relaxes the proportional odds assumption by allowing some predictors (LagDose, LagHb, Month and their interactions) to have non-proportional effects across the ordinal categories while maintaining proportional effects for others (sex, age and iron status).

The partial proportional odds model for a specific subject i at time j for threshold k can be specified as follows:

$$
\begin{aligned}
E\left[\log\left(\frac{P(Y_{ij} \leq k)}{1 - P(Y_{ij} \leq k)}\right)\right] &= \log\left(\frac{\pi_{jk}}{1 - \pi_{jk}}\right) \\
&= \beta_{0k} + \beta_{1k} \cdot \text{Month}_{ij} + \beta_{2k} \cdot \text{LagDose}_i + \beta_{3k} \cdot \text{LagHb}_i \\
&\quad + \beta_4 \cdot \text{Iron}_i + \beta_5 \cdot \text{Sex}_i + \beta_6 \cdot \text{Age}_i \\
&\quad + \beta_{7k} \cdot \text{LagDose}_i \times \text{Month}_{ij} + \beta_{8k} \cdot \text{LagHb}_i \times \text{Month}_{ij} \\
&\quad + \beta_{9k} \cdot \text{LagDose}_i \times \text{Iron}_i + \beta_{10k} \cdot \text{LagHb}_i \times \text{Iron}_i \\
&\quad + \beta_{11k} \cdot \text{LagDose}_i \times \text{LagHb}_i + \beta_{12k} \cdot \text{Month}_{ij} \times \text{Age}_i \\
&\quad + \beta_{13k} \cdot \text{Month}_{ij} \times \text{Iron}_i + \beta_{14k} \cdot \text{Month}_{ij} \times \text{Sex}_i \\
&\quad + \beta_{15k} \cdot \text{LagDose}_i^2 \\
&\quad + \beta_{16k} \cdot \text{LagHb}_i^2
\end{aligned}
$$

For $i = 1, \ldots, n$ subjects, measured across $j = 1, \ldots, 6$ time points for each subject i, and $k = 1, 2$ representing the two cumulative Hb category splits (Severe Anemia vs. Mild/Normal and Severe/Mild Anemia vs. Normal Hemoglobin)

- π_{jk}: The cumulative probability of being in a hemoglobin category less than or equal to k at time j.
- β_{0k}: The intercept term for cumulative category k, representing the baseline log-odds of being in the reference category (Severe Anemia) relative to the other categories (Mild/Normal Hb).
- β_{1k}: The effect of time (Month) on the cumulative log-odds of being in a given Hb category relative to the next (for each threshold k).

- β_{2k}: The effect of LagDose (previous EPO dose) on the cumulative log-odds of Hb category transitions for each threshold k.
- β_{3k}: The effect of LagHb (previous hemoglobin levels) on the cumulative log-odds of Hb category transitions for each threshold k.
- β_4: The effect of Iron status (binary: 0 = deficient, 1 = adequate) on the cumulative log-odds of transitioning between Hb categories.
- β_5: The effect of Sex (binary: 0 = male, 1 = female) on the cumulative log-odds of Hb category transitions.
- β_6: The effect of Age on the cumulative log-odds of Hb category transitions.
- β_{7k}: The interaction between LagDose and Month, indicating how the effect of previous EPO dose varies over time for each threshold k.
- β_{8k}: The interaction between LagHb and Month, indicating how the effect of previous hemoglobin levels varies over time for each threshold k.
- β_{9k}: The interaction between LagDose and Iron, indicating how the effect of EPO dose varies by iron status for each threshold k.
- β_{10k}: The interaction between LagHb and Iron, indicating how the effect of previous hemoglobin levels varies by iron status for each threshold k.
- β_{11k}: The interaction between LagDose and LagHb, indicating how the effects of both previous EPO dose and hemoglobin levels interact for each threshold k.
- β_{12k}: The interaction between Month and Age, indicating how the effect of age varies over time for each threshold k.
- β_{13k}: The interaction between Month and Iron, indicating how the effect of Iron status varies over time for each threshold k.
- β_{14k}: The interaction between Month and Sex, indicating how the effect of sex varies over time for each threshold k.
- β_{15k}: The quadratic effect of LagDose on the cumulative log-odds of Hb category transitions for each cumulative threshold k.
- β_{16k}: The quadratic effect of LagHb on the cumulative log-odds of Hb category transitions for each cumulative threshold k.

Note: The above-mentioned β's represent the population-level estimates for the effects of the covariates on the transitions between hemoglobin categories.

This model considers 17 parameters to describe the mean structure. The exchangeable (EXCH) working correlation structure was chosen to model within-subject correlation, assuming constant correlation across repeated measures of hemoglobin status. This structure is computationally efficient, robust to model misspecifications, and well-suited for unbalanced datasets with variable follow-up times and dropout, effectively accounting for within-subject dependence in longitudinal data.

The Partial Proportional Odds Model (PPOM) was implemented in SAS using the `GENMOD` procedure. Since `GENMOD` does not inherently support partial proportional odds models for multinomial outcomes, an alternative parameterization was employed. This involved transforming the ordinal response variable (hemoglobin categories: severe anemia, mild anemia, and normal hemoglobin) into a series of binary outcomes based on cumulative thresholds:

- $Y_{ij,k=1}$: Severe Anemia ($\leq k = 1$) vs. Mild/Normal,
- $Y_{ij,k=2}$: Severe/Mild ($\leq k = 2$) vs. Normal Hemoglobin.

Each cumulative threshold represents a split point that separates the categories into two groups, and a split variable (k) was introduced to distinguish between these cumulative comparisons. The split variable interacts with the chosen predictors, allowing for threshold-specific effects. For instance:

- **For** $k = 1$: The binary outcome represents the log-odds of being in Severe Anemia ($\leq k = 1$) compared to Mild/Normal. For this threshold, $\beta_{1k} \cdot \text{Month}_{ij}$ captures how time influences transitions from Severe Anemia to higher categories, reflecting either improvement or persistence over time.
- **For** $k = 2$: The binary outcome represents the log-odds of being in Severe/Mild Anemia ($\leq k = 2$) compared to Normal Hemoglobin. For this threshold, $\beta_{1k} \cdot \text{Month}_{ij}$ reflects how time affects transitions from Severe/Mild Anemia to Normal Hemoglobin.

This parameterization enabled the model to relax the proportional odds assumption by allowing non-proportional predictors such as LagDose, LagHb, and Month to have unique effects at each threshold. Meanwhile, covariates such as Iron, Sex, and Age were modeled with proportional effects across all thresholds.

3.3 Generalized Linear Mixed Models (GLMM)

GLMMs extend the Generalized Linear Model (GLM) framework by incorporating random effects to model subject-specific variability. This extension is particularly suited for longitudinal and hierarchical data, where observations are correlated within subjects, and the relationship between predictors and the response is modeled through both fixed and random effects.

For the partial proportional odds model it is specified as:

$$\text{logit}(P(Y \leq k)) = \alpha_k + \sum_{j=1}^{p_1} X_{1j}\beta_j + \sum_{j=1}^{p_2} X_{2j}\beta_{jk} + b_u + b_v \cdot \text{Time}, \quad k = 1, \ldots, K-1,$$

where:

- $\sum_{j=1}^{p_1} X_{1j}\beta_j$: Contribution of fixed effects for proportional predictors (X_1).
- $\sum_{j=1}^{p_2} X_{2j}\beta_{jk}$: Contribution of fixed effects for non-proportional predictors (X_2).
- b_u: Random intercept capturing subject-specific baseline deviations.

- $b_v \cdot$ Time: Random slope capturing variability in the effect of time across subjects.

Random Effects Distribution:

$$b = \begin{bmatrix} b_u \\ b_v \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_u^2 & \text{Cov}(u,v) \\ \text{Cov}(u,v) & \sigma_v^2 \end{bmatrix}\right),$$

where:

- σ_u^2: Variance of random intercepts.
- σ_v^2: Variance of random slopes.
- $\text{Cov}(u,v)$: Covariance between random intercepts and slopes.

The GLMM was estimated using the NLMIXED procedure, which allows for user-defined likelihood functions and therefore is adapted for the PPOLR model. The initial estimates were obtained from a GLIMMIX procedure and manually refined from the previous NLMIXED estimates after each run. Estimates were initially approximated using adaptative Gaussian quadrature to derive the mean structure with lower quadrature points (Q = 3) for computational efficiency. Likelihood Ratio Tests (LRT) were used to compare nested models and find the final mean structure.

The structure of random effects was examined by testing whether adding a random slope for time would significantly improve the model fit compared to a simpler model with only a random intercept and the result was evaluated based on the asymptotic null distribution, where -2$ln\lambda$ follows a $\chi_{1:2}$ distribution. Final estimates were obtained with adaptative Gaussian quadrature with higher points (Q = 10, 15) to ensure numerical stability and precision.

Empirical Bayes (EB) Predictions were obtained to estimate subject-specific predictions of random effects and used to examine if there are some deviating significantly from the population-level trends.

4 Results and Discussion

4.1 GEE

As observed during data exploration, non-proportional effects were identified for Month, LagDose, and LagHb, while proportional effects were observed for Age, Sex, and Iron Status. This indicates that the relationship between Month, LagDose, LagHb and their interactions and the log of probabilities for the two cumulative thresholds:

- $\text{logit } 1 = \log\left(\frac{P(Y_{ij} \text{ in Severe Anemia (category 1))}}{P(Y_{ij} \text{ in Mild/Normal (categories 2 or 3))}}\right)$,
- $\text{logit } 2 = \log\left(\frac{P(Y_{ij} \text{ in Severe/Mild Anemia (categories 1 or 2))}}{P(Y_{ij} \text{ in Normal Hemoglobin (category 3))}}\right)$,

Model selection was done using backward elimination based on the Quasi-likelihood under Independence Criterion (QIC). We started with a saturated model that included all main effects, interaction terms, and quadratic terms, using a binomial distribution

and a cumulative logit link function. Non-significant terms were removed one at a time, and QIC was recalculated after each step. A drop in QIC of more than 5 was taken as evidence of a better model fit. This process was repeated until we obtained the model with the lowest QIC.

Once the final model was selected, we tested different working correlation structures, including exchangeable (EXCH) (QIC =10839.6337), autoregressive (AR(1)) (QIC =10663.4879), and independence (IND) (QIC =10634.77). The independence structure had the lowest, indicating that the predictors included in the model effectively explained most of the variability between repeated measurements for the same subject. The results are summarized in Table 2.

The interaction between EPO dose and time indicates that higher cumulative doses of EPO over time are associated with increased log-odds of transitioning to improved hemoglobin categories ($\beta = 0.0010, p = 0.0450$). However, this effect diminishes for transitions to the highest hemoglobin category, where the interaction term is not significant ($\beta = -0.0001, p = 0.8094$), suggesting that the impact of EPO doses on hemoglobin stabilization may vary across clinical thresholds.

The interaction between prior hemoglobin levels and time shows that higher LagHb levels are associated with reduced log-odds of transitioning to higher hemoglobin categories ($\beta = -0.4603, p < 0.0001$ for earlier transitions; $\beta = -0.6270, p < 0.0001$ for transitions to normal levels). This suggests that higher baseline hemoglobin levels may reflect clinical stability, limiting further improvements as treatment progresses.

Moreover, the quadratic effect of prior hemoglobin levels was significant only when transitioning to the normal hemoglobin category ($\beta = -0.0482, p = 0.0099$), suggesting diminishing returns as baseline hemoglobin levels increase. This implies that as patients approach normal hemoglobin levels, the additional impact of higher baseline hemoglobin on further improvement decreases, reflecting a non-linear relationship. In contrast, the non-significant quadratic effect for transitions from severe to mild anemia ($\beta = 0.0008, p = 0.9660$) indicates a primarily linear relationship in this group, where increases in prior hemoglobin levels consistently contribute to improved outcomes without evidence of diminishing returns.

Iron deficiency and age did not exhibit significant temporal interactions, suggesting their influence on hemoglobin levels remained stable throughout the study. The fixed effect of sex on Hb categories was not significant. Of note, we did not include temporal interactions with sex due to limited biological evidence to suggest that sex would play a time-dependent role in hemoglobin dynamics in this context.

Final models are written as follows:

$$\text{Logit 1} = \log\left(\frac{P(Y_{ij} \text{ in Severe Anemia (HbCat = 1)})}{P(Y_{ij} \text{ in Mild/Normal (HbCat = 2 or 3)})}\right)$$

$$= -5.3717 + 4.5597 \cdot \text{Month}_{ij} - 0.0040 \cdot \text{LagDose}_i + 0.5060 \cdot \text{LagHb}_i + 0.0008 \cdot \text{LagHb}_i^2$$

$$+0.0010 \cdot \text{LagDose}_i \cdot \text{Month}_{ij} - 0.4603 \cdot \text{LagHb}_i \cdot \text{Month}_{ij} - 0.3138 \cdot \text{Iron}_i - 0.0869 \cdot \text{Sex}_i - 0.0066 \cdot \text{Age}_i.$$

$$\text{Logit 2} = \log\left(\frac{P(Y_{ij} \text{ in Severe/Mild Anemia (HbCat = 1 or 2)})}{P(Y_{ij} \text{ in Normal Hemoglobin (HbCat = 3)})}\right)$$

$$= -13.7300 + 7.0879 \cdot \text{Month}_{ij} - 0.0026 \cdot \text{LagDose}_i + 1.9349 \cdot \text{LagHb}_i - 0.0482 \cdot \text{LagHb}_i^2$$

$$-0.0001 \cdot \text{LagDose}_i \cdot \text{Month}_{ij} - 0.6270 \cdot \text{LagHb}_i \cdot \text{Month}_{ij} - 0.3138 \cdot \text{Iron}_i - 0.0869 \cdot \text{Sex}_i - 0.0066 \cdot \text{Age}_i.$$

Table 2: Analysis of GEE Parameter Estimates (Final Model)

Parameter	**Estimate**	**Standard Error**	**Lower 95% CI**	**Upper 95% CI**	**Z**	**P > —Z—**
Intercept	0.0000	0.0000	0.0000	0.0000	0.00	.
split 1	-5.3717	2.7047	-10.6729	-0.0706	-1.99	0.0470*
split 2	-13.7300	2.7560	-19.1317	-8.3283	-4.98	¡0.0001*
Age	-0.0066	0.0064	-0.0191	0.0059	-1.03	0.3034
iron (adequate)	-0.3138	0.2074	-0.7202	0.0926	-1.51	0.1302
sex (male)	-0.0869	0.0555	-0.1957	0.0219	-1.57	0.1174
month*split 1	4.5597	0.5610	3.4602	5.6591	8.13	< 0.0001*
month*split 2	7.0879	0.6833	5.7487	8.4271	10.37	< 0.0001*
lagdose*split 1	-0.0040	0.0014	-0.0069	-0.0012	-2.77	0.0056*
lagdose*split 2	-0.0026	0.0014	-0.0055	0.0002	1.83	0.0678
laghb*split 1	0.5060	0.4396	-0.3556	1.3675	-1.15	0.2497
laghb*split 2	1.9349	0.4306	1.0910	2.7788	-4.49	< 0.0001*
laghb*laghb*split 1	0.0008	0.0190	-0.0364	0.0380	-0.04	0.9660
laghb*laghb*split 2	-0.0482	0.0187	-0.0849	-0.0116	-2.58	0.0099*
month*lagdose*split 1	0.0010	0.0005	0.0000	0.0020	2.00	0.0450*
month*lagdose*split 2	-0.0001	0.0005	-0.0011	0.0009	-0.24	0.8094
month*laghb*split 1	-0.4603	0.0515	-0.5612	-0.3594	-8.94	¡0.0001*
month*laghb*split 2	-0.6270	0.0560	-0.7368	-0.5173	-11.20	< 0.0001*
Age*month*split 1	0.0014	0.0023	-0.0030	0.0059	0.64	0.5223
Age*month*split 2	0.0025	0.0022	-0.0019	0.0068	1.11	0.2689
month*iron*split 1	0.0936	0.0731	-0.0496	0.2367	1.28	0.2002
month*iron*split 2	0.0399	0.0730	-0.1031	0.1829	0.55	0.5844

4.2 GLMM

4.2.1 Test for Random Slopes

To determine the best random effects structure, we tested whether adding a random slope for time improved the model fit compared to a simpler model with only a random intercept. The p-value obtained was $p = 0.5 \times 0 + 0.5 \times P(\chi^2_1 > 27) \approx 0$, indicating significant improvement with the addition of a random slope. Therefore, the random intercept and random slope model was considered. After the mean structure reduction the final generalized linear mixed model was defined as follows:

$$\begin{aligned}\log\left(\frac{P(\text{HBCAT}_{ij} \leq k)}{1 - P(\text{HBCAT}_{ij} \leq k)}\right) &= \beta_k + \beta_{1k} \cdot \text{Month}_{ij} + \beta_{2k} \cdot \text{Age}_i \\ &+ \beta_{3k} \cdot \text{Sex}_i + \beta_{4k} \cdot \text{Iron}_i + \beta_{5k} \cdot \text{LagHB}_{ij} + \beta_{6k} \cdot (\text{Month}_{ij} \cdot \text{LagHB}_{ij}) \\ &+ \beta_{7k} \cdot (\text{LagDose}_{ij} \cdot \text{Iron}_i) + \beta_{8k} \cdot (\text{Month}_{ij} \cdot \text{Age}_i) \\ &+ \beta_{9k} \cdot (\text{Month}_{ij} \cdot \text{Sex}_i) + \beta_{10k} \cdot (\text{Age}_i \cdot \text{Sex}_i) + b_{1i}. \quad (1)\end{aligned}$$

for $i = 1, \ldots, n$ subjects, across $j = 1, \ldots, T$ time points measured for subject i, and $k = 1, 2$ threshold levels separating the cumulative categories of hemoglobin levels (HBCAT).

- $P(\text{HBCAT}_{ij} \leq k)$: The cumulative probability of being in HBCAT category k or lower at time j for subject i.
- β_k: The estimate for the intercept for cumulative category k.
- β_{1k}: The estimate for the time effect of **Month** for cumulative category k.
- β_{2k}: The estimate for the effect of **Age** for cumulative category k.
- β_{3k}: The estimate for the effect of **Sex** for cumulative category k, where

$$\text{Sex}_i = \begin{cases} 1 & \text{if subject } i \text{ is male,} \\ 2 & \text{if subject } i \text{ is female.} \end{cases}$$

- β_{4k}: The estimate for the effect of **Iron** for cumulative category k, where

$$\text{Iron}_i = \begin{cases} 0 & \text{if subject } i \text{ has inadequate iron stores,} \\ 1 & \text{if subject } i \text{ has adequate iron stores.} \end{cases}$$

- β_{5k}: The estimate for the effect of **LagHB** for cumulative category k.
- β_{6k}: The estimate for the interaction effect between **Month** and **LagHB** for cumulative category k.
- β_{7k}: The estimate for the interaction effect between **LagDose** and **Iron** for cumulative category k.
- β_{8k}: The estimate for the interaction effect between **Month** and **Age** for cumulative category k.

- β_{9k}: The estimate for the interaction effect between **Month** and **Sex** for cumulative category k.
- β_{10k}: The estimate for the interaction effect between **Age** and **Sex** for cumulative category k.
- b_{1i}: The random intercept for subject i, accounting for between-subject variability in baseline odds, assumed to follow $b_{1i} \sim N(0, \sigma_u^2)$.

Table 3: Test for Random Slope

Model	**-2log lik**	**Chisq**	**p-value**
Random intercept	10046	-	-
Random intercept + slope	10019	27	<.0001

4.2.2 Final Model Estimates

Table 4: Analysis of GLMM Parameter Estimates

Parameter	**Estimate**	**SE**	**P value**	**95% CI LB**	**95% CI UB**
Intercept1	1.4106	1.2175	0.2467	-0.9767	3.7979
Intercept2	7.7130	1.2911	<.0001	5.1815	10.2446
Age	-0.00982	0.007313	0.1794	-0.02416	0.004518
Sex_Men	0.4434	0.2200	0.0439	0.01204	0.8748
Iron_Inadequate	-0.4331	0.2505	0.0839	-0.9242	0.05797
Lag Dose	-0.00918	0.003406	0.0071	-0.01586	-0.00250
Partial Lag Dose	-0.00317	0.000740	<.0001	-0.00462	-0.00172
Lag HB	-0.2553	0.1013	0.0118	-0.4541	-0.05662
Partial Lag HB	-0.1139	0.05738	0.0472	-0.2264	-0.00140
Month	1.4812	0.3928	0.0002	0.7109	2.2514
Partial Month	-0.00589	0.1170	0.9599	-0.2353	0.2236
Lag HB*Month	-0.1563	0.03274	<.0001	-0.2205	-0.09214
Lag Dose*Iron	-0.00177	0.000923	0.0549	-0.00358	0.000037
Lag Dose*Lag HB	0.000851	0.000318	0.0075	0.000227	0.001474
Age*Month	0.002770	0.002389	0.2464	-0.00191	0.007454
Month*Iron_0	0.1417	0.07539	0.0602	-0.00611	0.2895
Month*Sex_0	-0.08572	0.07162	0.2314	-0.2262	0.05471
Random Intercept	3.1235	0.3116	<.0001	2.5126	3.7344
Random Slope	0.7186	0.09116	<.0001	0.5399	0.8974
Covariance Intercept-Slope	-1.8775	0.4703	<.0001	-2.7997	-0.9552

The GLMM results in table 4 provide insights into the dynamics of hemoglobin improvement over time. For every additional month, a 1-unit increase in LagHB is associated with a 14.5% decrease in the odds of transitioning to a higher hemoglobin category ($\beta = -0.1563$, $p < 0.0001$), indicating that higher prior hemoglobin levels reduce

the likelihood of improvement over time. Similarly, being male is associated with a non-significant 8.2% decrease in the odds of transitioning to a higher hemoglobin category for every additional month ($\beta = -0.08572$, $p = 0.2314$), suggesting no strong evidence of a sex-based effect on hemoglobin improvement over time. Conversely, being in the adequate iron group is associated with a 15.2% increase in the odds of transitioning to a higher hemoglobin category for every additional month ($\beta = 0.1417$, $p = 0.0602$), reflecting a marginally significant positive effect. Finally, a 1-year increase in age is associated with a non-significant 0.28% increase in the odds of transitioning to a higher hemoglobin category for every additional month ($\beta = 0.00277$, $p = 0.2464$), indicating no strong evidence that age influences hemoglobin improvement over time.

Table 5: Covariance Parameter Estimates for GLMM

Covariance Parameter	**Estimate**	**SE**
d11	9.7561	1.9463
d22	0.5164	0.1310
d12	-1.8775	0.4703

The variance and covariance estimates of the GLMM model are presented in Table 5. The results highlight high variability between subjects, as indicated by a random intercept variance d11 of 9.756 ($SE = 1.946$). In contrast, the variability within subjects over time, represented by the random slope variance d22, is lower at 0.516 ($SE = 0.131$). Additionally, the estimated covariance d12 of -1.877 ($SE = 0.470$) indicates an inverse relationship between the random intercept and the random slope. This suggests that subjects with higher initial HBCAT values tend to exhibit less improvement in their hemoglobin levels over time.

4.2.3 Empirical Bayes Estimates

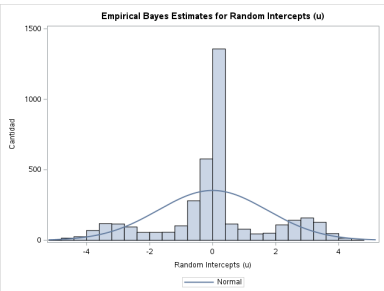

Figure 11: Histogram of Random Intercepts

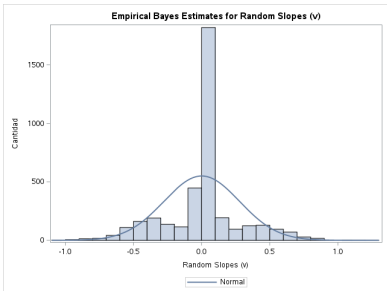

Figure 12: Histogram of Random Slopes

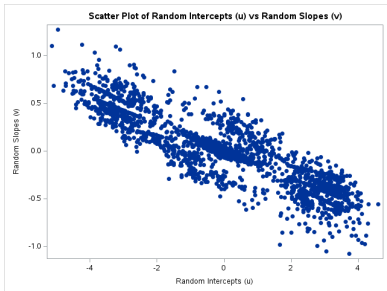

Figure 13: Scatter Plot of Random Intercepts and Slopes

The histograms for random intercepts suggest most individuals have Hb levels relatively close to the population mean, however, there are two groups with unusually high and low levels of starting Hemoglobin, expected by clinical differences. The histogram of random slopes also shows a similar effect, where most patients respond similarly to the main population but there are two distinguished groups of unusually high or low

responders to treatment or with different rates of progression in their anemic condition. Each point in the scatter plot represents an individual subject's EB estimate for the random intercept and slope. We can see a clear negative correlation between them, with patients starting from worse conditions improving faster and those starting in better conditions remaining more stable, still allowing for some individual variability as shown by the spread. This is consistent with what was observed in the population-level covariance structure estimates, but these estimates now allow for subject-specific predictions while accounting for the shrinkage effect towards the population mean.

5 GEE vs GLMMs

From a clinical perspective, the differences in results between GLMM and GEE emphasize the nuanced interpretations of how previous dose, sex, age, previous Hb levels and iron deficiency influence hemoglobin dynamics over time. In the GEE, LagDose had a modest but significant positive effect in transitions out of Severe Anemia (Split 1), reflecting the expected population-level role of EPO treatment in improving hemoglobin levels over time. However, this effect was not significant for transitions to Normal Hemoglobin (Split 2), suggesting that EPO's immediate impact may primarily support initial improvements, with other factors playing a larger role in achieving normal hemoglobin levels. This distinction is absent in the GLMM, which does not highlight LagDose's temporal effects.

Both models found no significant effect of sex or age on hemoglobin transitions over time, either at the population level (GEE) or within subject-specific trajectories (GLMM). This aligns with clinical understanding, as sex and age-based differences in hemoglobin dynamics may not be important in this context.

For Iron status the GLMM indicated a marginally significant positive effect of adequate iron status on hemoglobin improvement over time, suggesting that adequate iron supports better responses to EPO treatment. However, the GEE results did not show significant time-dependent interactions for iron status, implying that its effect is stable and not threshold-dependent. This discrepancy may arise because of the different modeling techniques and how they model the threshold-specific effects.

Prior Hb levels (LagHb) are significant in both models, were GLMM shows LagHb as a barrier to improvement, reducing the odds of transitioning to higher categories over time. GEE shows that higher LagHb increases the odds of remaining in more severe categories (Split 1 and Split 2), with a U-shaped relationship in Split 2 suggesting that patients with extreme LagHb levels may benefit more from treatment adjustments. This nuanced relationship is clinically relevant, reflecting the challenges of managing patients with moderate anemia levels compared to those at the extremes.

Overall, the GEE provides a broader population-level perspective on dose effects and threshold-specific dynamics, while the GLMM focuses on subject-specific trends, highlighting the predictors influence on individual improvement rates. These differences reflect the distinct ways each model captures clinical variability and temporal changes, offering complementary insights into hemoglobin management.

6 Conclusions

The exploratory analysis revealed substantial variability in hemoglobin transitions over time, driven by complex interactions among EPO dosing, LagHb, and iron status. Non-monotonic individual profiles highlight the dynamic and irregular nature of hemoglobin changes, reflecting nuanced clinical adjustments. Overlapping LagDose distributions across categories and the negative correlation between LagHb and LagDose suggest clinician-driven dosing decisions based on patient status. Furthermore, non-parallel cumulative log-odds for Month, LagDose, and LagHb indicate violations of the proportional odds assumption, reinforcing the need for sophisticated methods. Advanced approaches such as GEE, PPOLR, and GLMM (via NLMIXED) are essential to model population and individual-level effects, account for non-proportionality, and capture the irregular and interactive dynamics observed in this longitudinal data.

The GEE analysis provided critical population-level insights into the effects of key predictors on transitions between hemoglobin categories. LagDose showed a modest but significant positive effect for transitions out of Severe Anemia, indicating that higher EPO doses increase the odds of improvement in the most severe category. However, its limited effect on transitions to Normal Hemoglobin highlights that while dose escalation is effective in early-stage improvements, later transitions may depend more on factors such as iron status or long-term treatment adjustments. Similarly, LagHb demonstrated nuanced and clinically significant effects: higher prior hemoglobin levels were associated with increased odds of remaining in lower categories. This likely reflects clinician-driven treatment adjustments, where patients with moderate hemoglobin levels may experience dose reductions or maintenance strategies, leading to stabilization rather than further improvement. Importantly, GEE highlighted a U-shaped relationship for LagHb in transitions to Normal Hemoglobin, suggesting that extreme hemoglobin levels trigger dynamic treatment adjustments, while moderate levels are associated with stabilization rather than improvement. These findings emphasize the importance of dose intensification in Severe Anemia and the balance of stabilization and improvement at higher hemoglobin levels.

For iron status, the GLMM indicated a marginally significant positive effect of adequate iron status on hemoglobin improvement over time, suggesting that adequate iron supports better responses to EPO treatment. However, the GEE results did not show significant time-dependent interactions for iron status, implying that its effect is stable and not threshold-dependent. Both sex and age were found to have no significant effects on hemoglobin transitions over time by both models.

While GEE provides a robust framework for understanding population-averaged effects, it does not capture individual-level variability or the non-monotonic patterns observed in exploratory analyses. The observed interactions, particularly for LagHb and Month, suggest time-dependent effects that warrant a deeper exploration of individual trajectories. The GLMM approach allowed us to estimate subject-specific effects by incorporating random intercepts and slopes, accounting for the variability in individual responses. This is particularly valuable in understanding patient-level variability, such as why some patients respond to treatment more rapidly while others progress more slowly. The empirical Bayes estimates derived from the GLMM provided a deeper understanding of the distribution of individual responses and identified potential subpopulations that could benefit from tailored treatment regimens.

Overall, the combined use of GEE and GLMM underscored the complementary strengths

of these methods in addressing the complex dynamics of hemoglobin transitions. GEE excelled at capturing population-averaged effects and threshold-specific dynamics, while GLMM provided critical subject-specific insights and captured the variability in individual trajectories. Together, these models offered a comprehensive understanding of hemoglobin management, highlighting both the population-level effectiveness of EPO and the need for personalized approaches in treating anemia.

7 Appendix

```
/*Create lagged variables (lagdose, laghb)*/
data datasets.hemodialysis_lag;
    set datasets.hemodialysis;
    by ID MONTH;

    /* Initialize laghb and lagdose to missing */
    laghb = .;
    lagdose = .;

    /* For the first observation of each ID, leave laghb and lagdose as missing
*/
    if first.ID then do;
        laghb = .;  /* No lag for the first observation of the ID */
        lagdose = .; /* No lag for the first observation of the ID */
    end;
    else do;
        /* Calculate lag values, but handle cases where previous month is missing
*/
        if not missing(Hb) then laghb = lag(Hb);  /* Lag Hb only if the current
month's Hb is not missing */
        if not missing(DOSE) then lagdose = lag(DOSE);  /* Lag Dose only if
the current month's DOSE is not missing */

        /* Handle case where LAGDOSE is missing for the second month */
        if lagdose = . and not first.ID then lagdose = DOSE;  /* Assign the
current month's DOSE to lagdose */

        /* Handle case where LAGHB is missing for the second month */
        if laghb = . and not first.ID then lagHb = Hb;  /* Assign the current
month's Hb to laghb */
    end;

    /* Retain lag_Hb and lag_Dose across iterations */
    retain laghb lagdose;
run;

/* Creating HBCAT (new outcome) */
```

```
data datasets.hemodialysis_lag;
    set datasets.hemodialysis_lag;

    /* Create HBCAT based on Hb levels */
    if Hb <= 10 then HBCAT = 1;          /* Category 1: Hb  10 */
    else if 10 < Hb <= 12 then HBCAT = 2; /* Category 2: 10 < Hb  12 */
    else if Hb > 12 then HBCAT = 3;       /* Category 3: Hb > 12 */
    else HBCAT = .;                       /* Missing Hb results in missing
HBCAT */
run;

/*DATA EXPLORATION */
/*INDIVIDUAL PROFILES */
/* Create a random sample of patients */
proc surveyselect data=datasets.hemodialysis_lag out=datasets.sample_ids
                 method=srs
                 samprate=0.002
                 seed=54321;
    id ID;
run;

/* Keep only the sampled patients with their complete records */
data datasets.hemosample;
    merge datasets.hemodialysis_lag(in=a) datasets.sample_ids(in=b);
    by ID;
    if b;
run;

/*Plotting individual HBCAT profiles */
ods graphics / height=10in width=10in; /* Adjust size for multiple facets */

proc sgpanel data=datasets.hemosample;
    panelby ID / rows=4 columns=5 spacing=5; /* Adjusted for 20 profiles in
a single graph */
    series x=month y=HBCAT / lineattrs=(pattern=solid thickness=2);
    scatter x=month y=HBCAT / markerattrs=(symbol=circlefilled size=6);
    colaxis label="Month";
    rowaxis label="HBCAT (1=HB  10, 2=10 < HB  12, 3=HB > 12)";
    title "Individual Profiles of HBCAT for Each Patient";
run;

ods graphics / reset=all;

/*HBCAT PROPORTIONS OVER TIME */
proc sgplot data=datasets.hemodialysis_lag;
    vbar month / response=hbcat group=hbcat groupdisplay=stack;
    title "Proportion of each HB category Over Time";
    xaxis label="Month";
```

```
    yaxis label="Proportion of hbcat";
run;

/* Step 1: Count the number of patients in each HBCAT category per month */
proc sql;
    create table hbcat_counts as
    select month,
           hbcat,
           count(*) as count_hbcat
    from datasets.hemodialysis_lag
    group by month, hbcat;
quit;

/* Step 2: Calculate the total number of patients per month */
proc sql;
    create table hbcat_totals as
    select month,
           sum(count_hbcat) as total_count
    from hbcat_counts
    group by month;
quit;

/* Step 3: Merge the two datasets and calculate proportions */
data hbcat_proportions;
    merge hbcat_counts hbcat_totals;
    by month;
    proportion = (count_hbcat / total_count) * 100; /* Convert to percentage
*/
run;

/* Step 4: Plot the proportions */
proc sgplot data=hbcat_proportions;
    vbar month / response=proportion group=hbcat groupdisplay=stack;
    title "Proportion of each HB Category Over Time";
    xaxis label="Month";
    yaxis label="Proportion (%)";
run;


/*VARIANCE STRUCTURE */
/*Lagdose by month and HBCAT */
proc sgpanel data=datasets.hemodialysis_lag;
    panelby hbcat / rows=1 columns=3; /* One panel per HBCAT category */
    vbox lagdose / category=month fillattrs=(color=green transparency=0.3);
    colaxis label="Month";
    rowaxis label="LagDose";
    title "Distribution of LagDose by Month and HBCAT Category";
run;
```

```
proc sql;
    create table transitions as
    select id, month, lagdose, hbcat as current_hbcat,
           lag(hbcat) as previous_hbcat
    from datasets.hemodialysis_lag
    group by id
    having not missing(previous_hbcat);
quit;

proc sgplot data=transitions;
    vbox lagdose / category=previous_hbcat group=current_hbcat groupdisplay=cluster;
    title "LagDose Distribution by Transitions Between HBCAT Categories";
    xaxis label="Previous HBCAT Category";
    yaxis label="LagDose";
run;

/*Interaction?? lagodse*iron */
proc sgpanel data=datasets.hemodialysis_lag;
    panelby hbcat / rows=1 columns=3;
    vbox lagdose / category=iron fillattrs=(color=blue transparency=0.3);
    colaxis label="Iron Status (0=Deficient, 1=Adequate)";
    rowaxis label="LagDose";
    title "LagDose by Iron Status and HBCAT Categories";
run;

/*Interaction?? laghb*iron */
proc sgpanel data=datasets.hemodialysis_lag;
    panelby hbcat / rows=1 columns=3;
    vbox laghb / category=iron fillattrs=(color=blue transparency=0.3);
    colaxis label="Iron Status (0=Deficient, 1=Adequate)";
    rowaxis label="LagHb";
    title "LagHb by Iron Status and HBCAT Categories";
run;

/*Interaction?? laghb*lagdose */
proc sgpanel data=datasets.hemodialysis_lag;
    panelby hbcat / rows=1 columns=3;
    scatter x=laghb y=lagdose / group=hbcat markerattrs=(symbol=circlefilled
size=6);
    loess x=laghb y=lagdose / group=hbcat lineattrs=(pattern=solid thickness=2);
    colaxis label="LagHb (Previous Hemoglobin)";
    rowaxis label="LagDose (Previous EPO Dose)";
    title "LagHb vs. LagDose by HBCAT Categories with LOESS Smoothing";
run;

proc sgpanel data=datasets.hemodialysis_lag;
    panelby hbcat / rows=1 columns=3;
```

```
    scatter x=laghb y=lagdose / group=hbcat markerattrs=(symbol=circlefilled
size=6);
    reg x=laghb y=lagdose / group=hbcat lineattrs=(pattern=solid thickness=2);
    colaxis label="LagHb (Previous Hemoglobin)";
    rowaxis label="LagDose (Previous EPO Dose)";
    title "LagHb vs. LagDose by HBCAT Categories";
run;


/*Proportional odds assumption (PROC LOGISITC TO SEE SCORE TEST AND PLOT CUMLOGODDS)*

/*MONTH CONTINUOUS */
/* Step 1: Fit the Cumulative Logit Model */
proc logistic data=datasets.hemodialysis_lag descending;
    model HBCAT = Month / link=clogit;
    output out=logistic_predictions_month predicted=pred_prob;
run;

/* Step 2: Calculate Cumulative Probabilities and Logits */
data cumulative_probs_month;
    set logistic_predictions_month;
    retain cum_prob1 cum_prob2;

    /* Calculate cumulative probabilities */
    if HBCAT = 1 then cum_prob1 = pred_prob;
    else if HBCAT = 2 then cum_prob2 = cum_prob1 + pred_prob;

    /* Compute cumulative logits */
    if HBCAT = 1 and cum_prob1 > 0 and cum_prob1 < 1 then logit1 = log(cum_prob1
/ (1 - cum_prob1));
    if HBCAT = 2 and cum_prob2 > 0 and cum_prob2 < 1 then logit2 = log(cum_prob2
/ (1 - cum_prob2));
run;

/* Step 3: Aggregate Logits by Month */
proc means data=cumulative_probs_month noprint;
    class Month HBCAT;
    var logit1 logit2;
    output out=agg_logit_summary_month mean=mean_logit1 mean_logit2;
run;

/* Step 4: Plot Cumulative Log Odds by Month */
proc sgplot data=agg_logit_summary_month;
    series x=Month y=mean_logit1 / lineattrs=(pattern=solid) legendlabel="Logit
1 (HBCAT  1)";
    series x=Month y=mean_logit2 / lineattrs=(pattern=dash) legendlabel="Logit
2 (HBCAT  2)";
    title "Cumulative Log Odds by Month";
```

```
    xaxis label="Month" values=(1 to 6 by 1) grid;
    yaxis label="Cumulative Log Odds" grid;
run;

/* Lagdose */
proc logistic data=datasets.hemodialysis_lag descending;
    model hbcat = lagdose / link=clogit;
    output out=logistic_predictions predicted=pred_prob;
run;

data cumulative_probs_lagdose;
    set logistic_predictions;
    retain cum_prob1 cum_prob2;

    /* Calculate cumulative probabilities */
    if hbcat = 1 then cum_prob1 = pred_prob;
    else if hbcat = 2 then cum_prob2 = cum_prob1 + pred_prob;

    /* Compute cumulative logits */
    if hbcat = 1 and cum_prob1 > 0 and cum_prob1 < 1 then logit1 = log(cum_prob1
/ (1 - cum_prob1));
    if hbcat = 2 and cum_prob2 > 0 and cum_prob2 < 1 then logit2 = log(cum_prob2
/ (1 - cum_prob2));
run;

proc means data=cumulative_probs_lagdose noprint;
    class lagdose hbcat;
    var logit1 logit2;
    output out=agg_logit_summary_lagdose mean=mean_logit1 mean_logit2;
run;

proc sgplot data=agg_logit_summary_lagdose;
    series x=lagdose y=mean_logit1 / lineattrs=(pattern=solid) legendlabel="Logit
1 (HBCAT  1)";
    series x=lagdose y=mean_logit2 / lineattrs=(pattern=dash) legendlabel="Logit
2 (HBCAT  2)";
    title "Cumulative Log Odds by LagDose";
    xaxis label="LagDose" grid;
    yaxis label="Cumulative Log Odds" grid;
run;


/* LagHb */
/* Fit a proportional odds model with LAGHB as a continuous variable */
proc logistic data=datasets.hemodialysis_lag descending;
    model HBCAT = LAGHB / link=clogit; /* Proportional odds model */
    output out=logistic_predictions_laghb predicted=pred_prob;
run;
```

```
/* Calculate cumulative logits */
data cumulative_probs_laghb;
    set logistic_predictions_laghb;
    retain cum_prob1 cum_prob2;

    /* Calculate cumulative probabilities */
    if HBCAT = 1 then cum_prob1 = pred_prob;
    else if HBCAT = 2 then cum_prob2 = cum_prob1 + pred_prob;

    /* Compute cumulative logits */
    if HBCAT = 1 and cum_prob1 > 0 and cum_prob1 < 1 then logit1 = log(cum_prob1
/ (1 - cum_prob1));
    if HBCAT = 2 and cum_prob2 > 0 and cum_prob2 < 1 then logit2 = log(cum_prob2
/ (1 - cum_prob2));
run;

/* Aggregate cumulative logits */
proc means data=cumulative_probs_laghb noprint;
    class LAGHB HBCAT;
    var logit1 logit2;
    output out=agg_logit_summary_laghb mean=mean_logit1 mean_logit2;
run;

/* Plot cumulative log odds */
proc sgplot data=agg_logit_summary_laghb;
    series x=LAGHB y=mean_logit1 / lineattrs=(pattern=solid) legendlabel="Logit
1 (HBCAT  1)";
    series x=LAGHB y=mean_logit2 / lineattrs=(pattern=dash) legendlabel="Logit
2 (HBCAT  2)";
    title "Cumulative Log Odds by LagHb";
    xaxis label="LagHb" grid;
    yaxis label="Cumulative Log Odds" grid;
run;

/* SEX */
/* Fit a proportional odds model with SEX as a categorical variable */
proc logistic data=datasets.hemodialysis_lag descending;
    class sex_dummy (ref="0") / param=ref;
    model HBCAT = sex_dummy / link=clogit; /* Proportional odds model */
    output out=logistic_predictions_sex predicted=pred_prob;
run;

/* Calculate cumulative logits */
data cumulative_probs_sex;
    set logistic_predictions_sex;
    retain cum_prob1 cum_prob2;
```

```
    /* Calculate cumulative probabilities */
    if HBCAT = 1 then cum_prob1 = pred_prob;
    else if HBCAT = 2 then cum_prob2 = cum_prob1 + pred_prob;

    /* Compute cumulative logits */
    if HBCAT = 1 and cum_prob1 > 0 and cum_prob1 < 1 then logit1 = log(cum_prob1
/ (1 - cum_prob1));
    if HBCAT = 2 and cum_prob2 > 0 and cum_prob2 < 1 then logit2 = log(cum_prob2
/ (1 - cum_prob2));
run;

/* Aggregate cumulative logits */
proc means data=cumulative_probs_sex noprint;
    class sex_dummy HBCAT;
    var logit1 logit2;
    output out=agg_logit_summary_sex mean=mean_logit1 mean_logit2;
run;

/* Plot cumulative log odds */
proc sgplot data=agg_logit_summary_sex;
    series x=sex_dummy y=mean_logit1 / lineattrs=(pattern=solid) legendlabel="Logit
1 (HBCAT  1)";
    series x=sex_dummy y=mean_logit2 / lineattrs=(pattern=dash) legendlabel="Logit
2 (HBCAT  2)";
    title "Cumulative Log Odds by SEX";
    xaxis label="Sex (0 = Male, 1 = Female)" grid discreteorder=data;
    yaxis label="Cumulative Log Odds" grid;
run;


/* IRON */
/* Fit a proportional odds model with IRON as a categorical variable */
proc logistic data=datasets.hemodialysis_lag descending;
    class iron (ref="0") / param=ref;
    model HBCAT = iron / link=clogit; /* Proportional odds model */
    output out=logistic_predictions_iron predicted=pred_prob;
run;

/* Calculate cumulative logits */
data cumulative_probs_iron;
    set logistic_predictions_iron;
    retain cum_prob1 cum_prob2;

    /* Calculate cumulative probabilities */
    if HBCAT = 1 then cum_prob1 = pred_prob;
    else if HBCAT = 2 then cum_prob2 = cum_prob1 + pred_prob;

    /* Compute cumulative logits */
```

```
    if HBCAT = 1 and cum_prob1 > 0 and cum_prob1 < 1 then logit1 = log(cum_prob1
/ (1 - cum_prob1));
    if HBCAT = 2 and cum_prob2 > 0 and cum_prob2 < 1 then logit2 = log(cum_prob2
/ (1 - cum_prob2));
run;

/* Aggregate cumulative logits */
proc means data=cumulative_probs_iron noprint;
    class iron HBCAT;
    var logit1 logit2;
    output out=agg_logit_summary_iron mean=mean_logit1 mean_logit2;
run;

/* Plot cumulative log odds */
proc sgplot data=agg_logit_summary_iron;
    series x=iron y=mean_logit1 / lineattrs=(pattern=solid) legendlabel="Logit
1 (HBCAT  1)";
    series x=iron y=mean_logit2 / lineattrs=(pattern=dash) legendlabel="Logit
2 (HBCAT  2)";
    title "Cumulative Log Odds by IRON";
    xaxis label="Iron (0 = No Iron, 1 = Iron)" grid discreteorder=data;
    yaxis label="Cumulative Log Odds" grid;
run;


/* AGE */
/* Fit a proportional odds model with AGE as a continuous variable */
proc logistic data=datasets.hemodialysis_lag descending;
    model HBCAT = AGE / link=clogit; /* Proportional odds model */
    output out=logistic_predictions_age predicted=pred_prob;
run;

/* Calculate cumulative logits */
data cumulative_probs_age;
    set logistic_predictions_age;
    retain cum_prob1 cum_prob2;

    /* Calculate cumulative probabilities */
    if HBCAT = 1 then cum_prob1 = pred_prob;
    else if HBCAT = 2 then cum_prob2 = cum_prob1 + pred_prob;

    /* Compute cumulative logits */
    if HBCAT = 1 and cum_prob1 > 0 and cum_prob1 < 1 then logit1 = log(cum_prob1
/ (1 - cum_prob1));
    if HBCAT = 2 and cum_prob2 > 0 and cum_prob2 < 1 then logit2 = log(cum_prob2
/ (1 - cum_prob2));
run;
```

```
/* Aggregate cumulative logits */
proc means data=cumulative_probs_age noprint;
    class AGE HBCAT;
    var logit1 logit2;
    output out=agg_logit_summary_age mean=mean_logit1 mean_logit2;
run;

/* Plot cumulative log odds */
proc sgplot data=agg_logit_summary_age;
    series x=AGE y=mean_logit1 / lineattrs=(pattern=solid) legendlabel="Logit
1 (HBCAT  1)";
    series x=AGE y=mean_logit2 / lineattrs=(pattern=dash) legendlabel="Logit
2 (HBCAT  2)";
    title "Cumulative Log Odds by AGE";
    xaxis label="Age" grid;
    yaxis label="Cumulative Log Odds" grid;
run;

/*PROC LOGISTIC SCORE TEST FULL MODEL */
proc logistic data=datasets.hemodialysis_lag descending;
    model HBCAT = month lagdose laghb sex_dummy iron age lagdose*lagdose laghb*laghb
lagdose*month iron*month laghb*month lagdose*iron lagdose*laghb
                iron*laghb/
                link=clogit; /* Proportional odds model */
    output out=logistic_predictions_full predicted=pred_prob;
run;


/*PARTIAL PROPORTIONAL */

DATA hbcat_restructured;
    SET datasets.hemodialysis_lag; /* Original dataset */

    /* Create split levels for transitions */
    split = 1; y2 = (hbcat > 1); OUTPUT; /* Compare Severe (1) vs. Mild/Healthy
(2, 3) */
    split = 2; y2 = (hbcat > 2); OUTPUT; /* Compare Severe/Mild (1, 2) vs.
Healthy (3) */
RUN;

DATA hbcat_restructured;
    SET hbcat_restructured;
    month_class = month; /* Simply copy the numeric month variable */
RUN;

PROC SORT DATA=hbcat_restructured;
    BY id month_class; /* Ensure proper order within each subject */
RUN;
```

```
/* FULL SATURATED MODEL QIC =10841.1583 */
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*lagdose*lagdose split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*lagdose*iron split*laghb*iron split*lagdose*laghb
               split*month*age split*month*iron split*month*sex_dummy /
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=EXCH;
RUN;

/* MODEL 2 (NO SPLIT*LAGHB*IRON) QIC=10836.0233 */
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*lagdose*lagdose split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*lagdose*iron split*lagdose*laghb
               split*month*age split*month*iron split*month*sex_dummy /
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=EXCH;
RUN;

/* MODEL 3 (NO SPLIT*LADOSE*IRON) QIC = 10836.6240  */
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*lagdose*lagdose split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*lagdose*laghb
               split*month*age split*month*iron split*month*sex_dummy /
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=EXCH;
RUN;

/*MODEL 4 (NO SPLIT*LAGDOSE*LAGDOSE) QIC = 10835.2634 */
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*lagdose*laghb
```

```
                split*month*age split*month*iron split*month*sex_dummy /
                DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=EXCH;
RUN;

/*MODEL 5 (NO SPLIT*LAGDOSE*LAGHB) QIC = 10839.0952*/
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
                split*lagdose split*laghb
                split*laghb*laghb
                split*lagdose*month split*laghb*month
                split*month*age split*month*iron split*month*sex_dummy /
                DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=EXCH;
RUN;

/*MODEL 6 (NO SPLIT*MONTH*SEX) QIC = 10839.6337 */
PROC GENMOD DATA=hbcat_restructured  ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
                split*lagdose split*laghb
                split*laghb*laghb
                split*lagdose*month split*laghb*month
                split*month*age split*month*iron/
                DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=EXCH;
RUN;


/*WORKING CORR STRUCTURE (MONTH RELAXED) */

/*QIC 10835.4882 (IF KEEPING SPLIT*LAGDOSE*LAGHB BUT REMOVING SPLIT*MONTH*SEX)
*/
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
                split*lagdose split*laghb
                split*laghb*laghb
                split*lagdose*month split*laghb*month
                split*lagdose*laghb
                split*month*age split*month*iron/
                DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=EXCH;
RUN;

/* QIC= 10839.6337 MODEL 6 */
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
```

```
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*month*age split*month*iron/
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=EXCH;
RUN;

/*QIC 10662.5853 (IF KEEPING SPLIT*LAGDOSE*LAGHB BUT REMOVING SPLIT*MONTH*SEX)
*/
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*lagdose*laghb
               split*month*age split*month*iron/
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=AR(1);
RUN;

/*QIC 10663.4879 MODEL 6*/
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*month*age split*month*iron/
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=AR(1);
RUN;


/*QIC 10632.9465 (LOWEST) (IF KEEPING SPLIT*LAGDOSE*LAGHB BUT REMOVING SPLIT*MONTH*SE
*/
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*lagdose*laghb
               split*month*age split*month*iron/
               DIST=BINOMIAL LINK=LOGIT NOINT;
```

```
    REPEATED SUBJECT=id / TYPE=IND;
RUN;


/*QIC 10634.7677 MODEL 6*/
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*month*age split*month*iron/
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=IND;
RUN;

/* QIC WARNING: The number of response pairs for estimating correlation is
less than or equal to the number of regression parameters  */
PROC GENMOD DATA=hbcat_restructured DESCENDING ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*lagdose*laghb
               split*month*age split*month*iron/
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=UNSTR;
RUN;


/*FINAL PPO MODEL CONSIDERING LAGDOSE MONTH AND LAGHB NON PROP */
/*QIC 10634.7677 (LOWEST)*/
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*month*age split*month*iron/
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=IND;
RUN;

/* TABLES FOR REPORT */
/* FULL SATURATED MODEL QIC =10841.1583 */
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
```

```
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*lagdose*lagdose split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*lagdose*iron split*laghb*iron split*lagdose*laghb
               split*month*age split*month*iron split*month*sex_dummy /
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=EXCH;
RUN;

/*FINAL PPO MODEL CONSIDERING LAGDOSE MONTH AND LAGHB NON PROP */
/*QIC 10634.7677 (LOWEST)*/
PROC GENMOD DATA=hbcat_restructured ORDER=DATA;
    CLASS id split iron(ref="0") sex_dummy(ref="0") month_class;
    MODEL y2 = split age iron sex_dummy split*month
               split*lagdose split*laghb
               split*laghb*laghb
               split*lagdose*month split*laghb*month
               split*month*age split*month*iron/
               DIST=BINOMIAL LINK=LOGIT NOINT;
    REPEATED SUBJECT=id / TYPE=IND corrw MODELSE;
RUN;



/*GLIMMIX*/
proc glimmix data=hemo_data_lag;
   class ID SEX IRON HBCAT;
   model HBCAT = AGE SEX IRON lag_dose lag_hb month
                 /* Interactions */
                 lag_dose*month
                 lag_hb*month
                 lag_dose*iron
                 lag_hb*iron
                 lag_dose*lag_hb
                 month*age
                 month*iron
                 month*sex
                 / dist=multinomial link=cumlogit solution;
   random intercept / subject=ID type=vc;
   nloptions maxiter=200 technique=quanew;
   ods output ParameterEstimates=glimmix_params CovParms=glimmix_covparms FitStatisti
run;
/* FINAL MODEL WITH QPOINTS= 10 AND UPDATED VARIANCE PARAM */
proc nlmixed data=hemo_data_lag method=gauss qpoints=10 ;
  parms beta_intercept1=2.0781
       beta_intercept2=6.7498
       beta_age=-0.00987
```

```
   beta_sex_0=0.3242
   beta_iron_0=0.02225
   beta_lagdose=-0.00885
   beta_lagdose_p=-0.00267
   beta_laghb=-0.2581
   beta_laghb_p=-0.0892
   beta_month=1.2471
   beta_month_p=0.2461
   beta_laghb_month=-0.1469
   beta_lagdose_iron=-0.00163
   beta_lagdose_laghb=0.000806
   beta_age_month=0.002899
   beta_month_iron_0=0.1043
   beta_month_sex_0=-0.05500
   sigma_u=2.6431 sigma_v=0.5884  cov_uv=-1.2482;
eta1 = beta_intercept1
       + beta_age * AGE
       + beta_sex_0 * (SEX)
       + beta_iron_0 * (IRON)
       + beta_lagdose * lag_dose
       + beta_laghb * lag_hb
       + beta_month * month
       + beta_laghb_month * lag_hb * month
       + beta_lagdose_iron * lag_dose * IRON
       + beta_lagdose_laghb * lag_dose * lag_hb
       + beta_age_month * AGE * month
       + beta_month_iron_0 * (month * IRON)
       + beta_month_sex_0 * (month * SEX)
       + u + v * MONTH; /* Random intercept AND SLOPE */
eta2 = beta_intercept2
       + beta_age * AGE
       + beta_sex_0 * (SEX)
       + beta_iron_0 * (IRON)
       + (beta_lagdose + beta_lagdose_p) * lag_dose
       + (beta_laghb + beta_laghb_p) * lag_hb
       + (beta_month + beta_month_p) * month
       + beta_laghb_month * lag_hb * month
       + beta_lagdose_iron * lag_dose * IRON
       + beta_lagdose_laghb * lag_dose * lag_hb
       + beta_age_month * AGE * month
       + beta_month_iron_0 * (month * IRON)
       + beta_month_sex_0 * (month * SEX)
       + u + v*MONTH; /* Random intercept AND slope */
/* Cumulative probabilities */
p1 = exp(eta1) / (1 + exp(eta1));          /* P(HBCAT  1) */
p2 = exp(eta2) / (1 + exp(eta2)) - p1;    /* P(HBCAT = 2) */
p3 = 1 - p1 - p2;                         /* P(HBCAT = 3) */
```

```
    /* Log-likelihood */
    if HBCAT=1 then ll = log(p1);
    else if HBCAT=2 then ll = log(p2);
    else if HBCAT=3 then ll = log(p3);

    /* Model specification */
    model HBCAT ~ general(ll);

    /* Random effects */
    random u v ~ normal([0, 0], [sigma_u**2, cov_uv, sigma_v**2]) subject=ID
out=eb_predictions;

    estimate 'var(sigma_u)' sigma_u**2;
 estimate 'var(sigma_v)' sigma_v**2;
 estimate 'cov(u, v)' cov_uv;

    predict u out=predicted_intercepts;
    predict v out=predicted_slopes;

    ods output ConvergenceStatus=conv_status ParameterEstimates=params;
run;
```